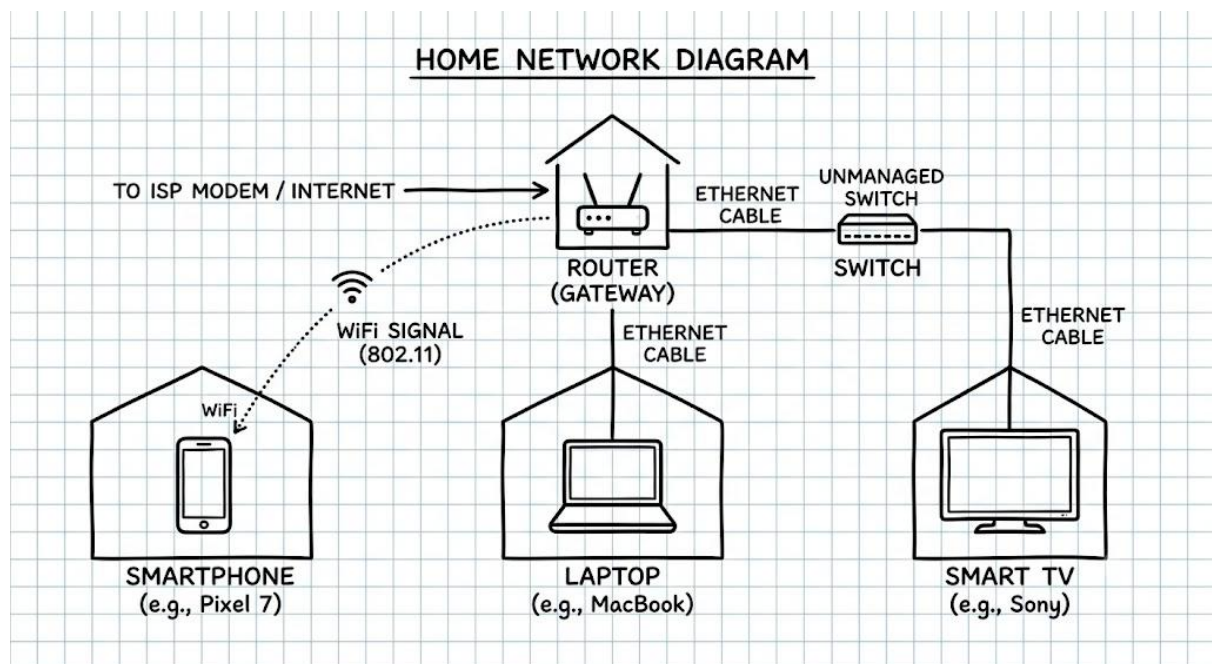


TASK-1



TASK-2

Network Device	Function	Real-World Use Case	Example App/Service
Router	Connects the local network to the Internet	Provides internet access to all home devices	YouTube streaming
Switch	Connects multiple wired devices in a LAN	Connects PCs for LAN gaming or office work	BGMI / Free Fire LAN setup
Hub	Broadcasts data to all connected devices	Used in small or old networks for basic connectivity	File sharing between PCs
Access Point (AP)	Provides wireless (Wi-Fi) access	Connects smartphones and laptops wirelessly	Instagram, WhatsApp, Netflix

TASK-3

Scenario: Ordering Food on Zomato

When I order food on **Zomato**, my **smartphone** connects to the **Access Point (Wi-Fi)**, which sends the request to the **Router**. The **Router** forwards the request to the Internet and the Zomato servers. If multiple wired devices are connected in the home network, a **Switch** transfers data between the router and those devices efficiently. After the order is processed, the confirmation travels back through the **Router**, **Access Point**, and finally reaches my phone.

TASK-4

Solution 1: WiFi Extender (with Ethernet Port)

- **How it works:** The WiFi extender connects wirelessly to the router and provides an **Ethernet port**. Connect the gaming PC to the extender using an Ethernet cable.
- **Benefit:** No long Ethernet cable is needed, and it extends the WiFi signal.

Solution 2: Powerline Adapter

- **How it works:** One powerline adapter connects to the router using an Ethernet cable, and the second adapter plugs into a power socket near the gaming PC. The PC connects to the second adapter using Ethernet.
- **Benefit:** Uses the home's electrical wiring to provide a stable and fast internet connection, which is ideal for gaming.